BQP · WISER SUMMER CHALLENGE 2026

Quantum-Assisted Physics-Informed Neural Networks for Aerospace Shock-Wave Modeling

A Mechanistic Investigation of Representation, Optimization and Spectral Bias

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Benchmark: Viscous Burgers' Equation · Physics-Informed Learning · Quantum Machine Learning

MOTIVATION

Why Computational Fluid Dynamics Matters

Aircraft	Hypersonics	Spacecraft	Gas Turbines	Defense
Shock-wave prediction	*Compressible flow*	*Re-entry aerodynamics*	*High-speed combustion*	*High-speed aerodynamics*

Shock-dominated, high-speed flow phenomena recur across every one of these domains.

Computational Fluid Dynamics (CFD)

↓

Traditional Numerical Solver

↓

High Computational Cost

↓

Physics-Informed Neural Networks (PINNs)

↓

Can Quantum Computing Improve Learning?

Framing the Research Problem

We are NOT trying to prove quantum is better.

Instead, we investigate:

01 Whether a quantum feature map reduces spectral bias

02 Whether it improves shock-region accuracy

03 If not fully realized, which architectural factors explain why

04 Which optimization strategies interact with representation

Research Objective

Evaluate whether a quantum data-encoding feature map can mitigate spectral bias in a hybrid QAPINN, relative to a classical PINN baseline, on the viscous Burgers' equation — and if the improvement does not fully materialize, identify which architectural and optimization factors explain why.

SCIENTIFIC QUESTIONS

Hypotheses Under Test

H1

Encoding affects learning.

H2

Increasing qubits improves approximation.

H3

Increasing depth improves learning.

H4

Entanglement improves performance.

H5

Optimizer interacts with representation.

H6

Increasing collocation improves learning.

H7

Lower PDE residual guarantees better prediction.

H8

A hybrid quantum layer can outperform an equivalent classical representation.

Model A vs. Model B — Implemented Architectures

Model A — Classical PINN

Input (x, t)

↓

Classical Dense Representation

↓

Hidden Layers (5 × 64, tanh)

↓

Output û(x, t)

Xavier-normal init · 16,897 trainable parameters

Model B — Quantum-Assisted PINN

Input (x, t)

↓

Normalization

↓

Quantum Encoding

↓

Parameterized Quantum Circuit

↓

Measurement

↓

Classical Readout

↓

Prediction û(x, t)

4 qubits · depth 3 · circular entanglement · 85 trainable parameters (≈0.50% of Model A)

Key design decision: unlike Model A, Model B's first representation layer is replaced directly by quantum encoding — no classical preprocessing dense layer precedes the quantum circuit.

Controlled Comparison Design

Burgers' PDE → Collocation Points (LHS) → **PINN / QAPINN** → PDE + IC + BC Loss → Adam → L-BFGS → Evaluation vs. Reference

Model A — Classical Dense Representation

Model B — Quantum Encoding + Circuit

Only this single representation stage differs — every other pipeline stage above is byte-for-byte identical.

Shared Across Both Models

- Reference Solution
- LHS Sampling
- Physics-Informed Loss
- Automatic Differentiation
- Training Schedule
- Evaluation Protocol

Experimental fairness: any measured difference is attributable to representation, not to a confounded pipeline.

Implemented QAPINN Pipeline (Model B)

End-to-End Pipeline

Input (x, t)

↓

Normalization

↓

Quantum Encoding

↓

Variational Quantum Circuit

↓

Measurement ⟨Z⟩

↓

Classical Readout (4→8→1)

↓

Prediction û(x, t)

85 trainable parameters total — ≈0.50% of Model A's 16,897

Zoom — Quantum Circuit Block

1. **Encoding (default: angle re-uploading)**
   (x,t) re-encoded via RY/RZ rotations before every variational layer
2. **Variational Rotations**
   Trainable single-qubit rotation gates — 4 qubits × 3 layers
3. **Entanglement (default: circular)**
   Two-qubit gates couple qubit correlations
4. **Measurement**
   ⟨Z⟩ expectation value read out on each qubit
5. **Classical Interface**
   Linear head 4→8→1 maps measurements to û(x,t)

PennyLane default.qubit simulator · backprop differentiation, verified against 2nd-order PDE derivatives

EXPERIMENTAL DESIGN

Single-Variable Controlled Ablation

CONTROLLED ABLATION STUDY

Variable	Role in Study	Baseline (Default)	Swept Range
Encoding	Data representation	Angle re-uploading	Angle · Angle re-uploading · Amplitude
Optimizer	Convergence strategy	Adam → L-BFGS	Adam-only vs. Adam + L-BFGS
Collocation Points	Sampling density	4,000 (Model B default)	800 · 2,000 · 4,000
Qubits	Circuit width	4	2 · 4 · 6
Circuit Depth	Circuit expressivity	3	1 · 2 · 4
Entanglement	Correlation structure	Circular	Linear · Circular · Full

Design rule: only ONE variable changes per experiment. Qubit / depth / entanglement sweeps ran at a reduced budget (800 pts, 300 Adam epochs, L-BFGS off) to make a 12-run sweep tractable.

Only Metrics Actually Measured in This Project

1

Prediction

- Relative L2 error (global)
- Shock-region L2 error
- RMSE

2

Physics

- PDE residual
- Initial-condition loss
- Boundary-condition loss

3

Spectral

- Fourier spectrum comparison
- High-frequency reconstruction

4

Efficiency

- Training time
- Parameter count
- Optimizer convergence

EXPERIMENTAL ROADMAP

Actual Research Chronology

01	02	03	04	05	06	07
Baseline	**Near-Trivial Diagnosis**	**Controlled Ablation**	**Optimizer × Representation**	**Collocation Density Study**	**Matched-Budget Validation**	**Final Comparative Analysis**
Establish Model A & Model B reference performance	Diagnose Model B's convergence toward u≡0	Sweep qubits, depth, entanglement, encoding	Controlled 2×2: encoding × L-BFGS on/off	Test 800–4,000 collocation points	Amplitude encoding at Model A's exact budget	Synthesize findings across all experiments

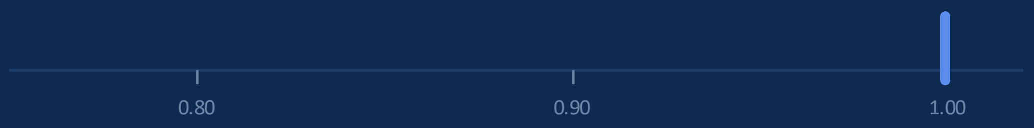
CONTROLLED ABLATION RESULTS

Global L2 Error Range Across Four Sweeps

Qubits

Values tested: 2, 4, 6

99.8% – 100.0%

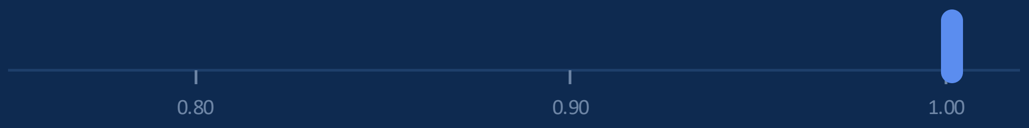


Flat — within shared plateau

Circuit Depth

Values tested: 1, 2, 4

99.9% – 100.5%

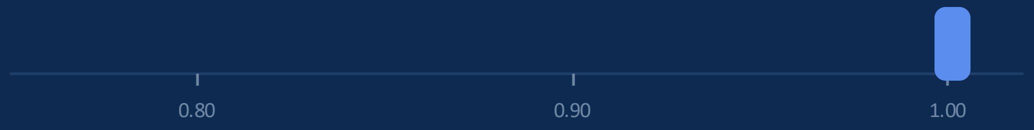


Flat — within shared plateau

Entanglement

Values tested: Linear, Circular, Full

99.0% – 100.0%



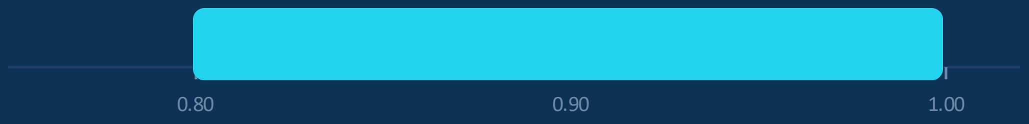
Flat — within shared plateau

Encoding

KEY FACTOR

Values tested: Angle, Angle re-upl., Amplitude

80.0% – 100.0%



Signal — amplitude partially escapes the plateau

KEY SCIENTIFIC FINDING

Optimizer × Representation Interaction

Angle re-uploading

Adam-only

Global L2: **100.04%**

Shock L2: **100.03%**

Angle re-uploading

Adam + L-BFGS

Global L2: **98.35%**

Shock L2: **100.03%**

Amplitude

Adam-only

Global L2: **79.96%**

Shock L2: **98.64%**

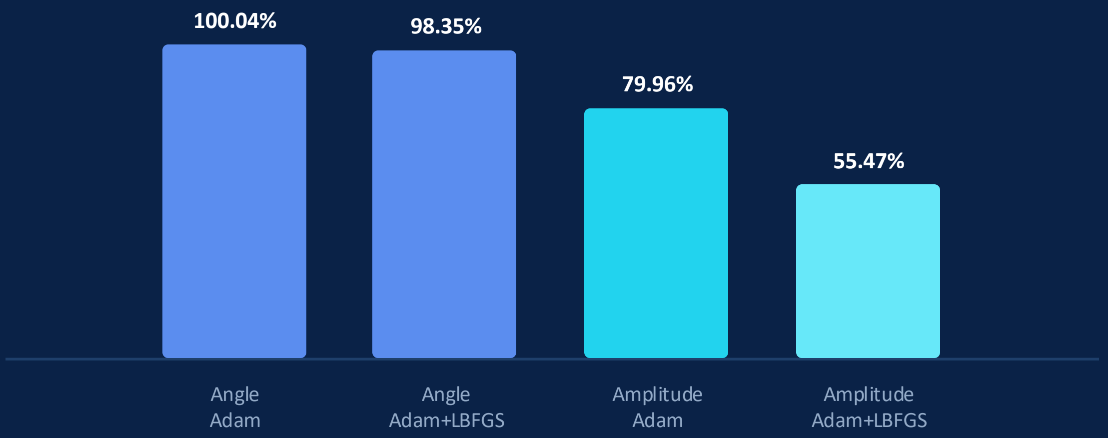
Amplitude

Adam + L-BFGS

Global L2: **55.47%**

Shock L2: **89.12%**

Global L2 Error by Configuration



Encoding	Adam-only L2	Adam+L-BFGS L2	Shock L2 (Adam+LBFGS)
Angle re-uploading	100.04%	98.35%	100.03% (unchanged)
Amplitude	79.96%	55.47%	89.12%

Representation determines optimization effectiveness — the optimizer that helps depends on how data is encoded.

UNEXPECTED FINDINGS

Results That Diverged From Theory

Finding 1

Increasing collocation density produced almost no improvement — global L2 error stayed within 55–58% across an 800 → 4,000 point, 5× range.

Finding 2

Lower PDE residual did not consistently correspond to lower prediction error — residual and accuracy moved together in some runs and traded off in others.

Finding 3

Theoretical analyses often motivate angle-based encodings, whereas amplitude encoding performed best in our experiments.

OPEN RESEARCH QUESTION

DISCUSSION

Hypothesis-by-Hypothesis Verdict

#	Hypothesis	Verdict
H1	Encoding affects learning.	✓ Supported
H2	Increasing qubits improves approximation.	✗ Rejected
H3	Increasing depth improves learning.	✗ Rejected
H4	Entanglement improves performance.	✗ Rejected
H5	Optimizer interacts with representation.	✓ Supported
H6	Increasing collocation improves learning.	✗ Rejected
H7	Lower PDE residual guarantees better prediction.	◐ Partially Supported
H8	A hybrid quantum layer can outperform an equivalent classical representation.	✗ Rejected

✓ Supported

◐ Partially Supported

✗ Rejected

CONCLUSION

Answering the Research Question

Configuration	L2 (global)	L2 (shock)	Params
Model A — Classical PINN	4.01%	11.83%	16,897
Model B — baseline (angle re-upl.)	98.36%	99.85%	85
Model B — amplitude + L-BFGS (matched budget)	**47.02%**	**78.79%**	**85**

Gap to classical narrowed from ≈24× (baseline) to ≈11.7× (best result) — through representation and optimization alone, with parameter count fixed.

Main Findings

- Encoding, not circuit size, determined performance.
- Optimizer effectiveness depended on representation.
- All gains came from representation and training strategy — parameter count never changed.
- Classical PINN remained the stronger performer throughout.

Future Work

- Real quantum hardware validation (IBM Quantum)
- Extend to heat/wave equation and 2-D Navier–Stokes
- Adaptive, problem-specific feature maps
- Isolate expressivity vs. trainability behind amplitude encoding's advantage

KEY TAKEAWAY

Representation governs learning more than circuit size. This work identifies when and why quantum representations influence PINN learning — rather than claiming an unconditional quantum advantage.

REPRODUCIBILITY

Environment and Configuration

Item	Value
Random seed	42 (fixed across all runs — config.py / config_b.py)
Python version	3.13.7
PyTorch version	2.13.0 (CPU)
PennyLane version	0.45.1
Operating system	Microsoft Windows 11 Home (Build 26200)
CPU / RAM	13th Gen Intel Core i7-1355U / 16 GB
Quantum backend	PennyLane default.qubit (noiseless simulator), backprop differentiation
GitHub repository	GitHub - meenakshi-re18/qapinn-bqp-challenge · GitHub

Notes

- All experiments are runnable end-to-end from the project's public GitHub repository.
- All reported training times are wall-clock CPU times; no GPU acceleration was used for either model.

Thank You · Questions?